Jiayu Tang

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RESEARCH INTERESTS

Computational + experimental antibody research: understanding and predicting how antibody context (framework, CDR loops, VH–VL pairing, and epitope geometry) shapes antigen recognition, and testing when AI structure-prediction and design tools can be trusted, with the goal of informing antibody design and engineering. Long term, I aim to make biological interactions predictable and engineerable by integrating computational discovery, wet-lab validation, and statistical interpretation.

EDUCATION

University of Nebraska–Lincoln, Lincoln, NE 08/2024 – 08/2026

M.S., Food Science and Technology; Minor in Statistics · GPA 3.95/4.00 · Advisor: Dr. Edward Deehan

Graduate statistics coursework: Statistical Methods in Research (STAT 801A), Design and Analysis of Research Studies (STAT 802), Multiple Regression (STAT 870), Applied Multivariate Statistics (STAT 873)

University of Nebraska–Lincoln, Lincoln, NE 08/2023 – 05/2024

B.Sc., Food Science and Technology ("3+1" dual-degree program)

Northwest A&F University, Yangling, China 09/2020 – 06/2024

B.Eng., Food Science and Engineering ("3+1" dual-degree program) · GPA 3.52/4.00

RESEARCH EXPERIENCE

OPS Research Assistant (full-time), Wang Lab 10/2026 – Present

Department of Infectious Diseases and Immunology, College of Veterinary Medicine, University of Florida · Supervisor: Dr. Yiquan Wang

- Investigating how human VH–VL framework architecture influences predicted antibody–antigen recognition under fixed CDR, antigen, and epitope controls.
- Built an in-silico antibody-design pipeline using framework grafting, native controls, and epitope-masked structural evaluation.
- Ran large-scale AlphaFold/ColabFold antibody–antigen predictions and evaluated model confidence, docking accuracy, and interface contacts.
- Benchmarked emerging antibody-design tools, including RFantibody and Germinal.
- Used AI coding agents (Claude Code, Codex) to plan, implement, and audit computational workflows.

Graduate Research Assistant, Deehan Lab 07/2024 – 07/2026

Department of Food Science and Technology, University of Nebraska–Lincoln · Advisor: Dr. Edward Deehan

Randomized crossover clinical trial: diet–gut microbiome–host interactions

- Coordinated a 12-week, three-phase randomized crossover trial of dry bean market-class effects on gut microbiota and cardiometabolic health markers (IRB-approved; registered on ClinicalTrials.gov).
- Co-developed the NIH-style protocol, informed consent, and recruitment materials with the PI; authored 4 SOPs and 2 clinical-trial workflows.
- Took over trial coordination after the clinical research coordinator resigned; led 200+ clinic visits and managed recruitment, scheduling, documentation, and supplies.
- Processed human blood (centrifugation, plasma/serum separation, aliquoting) and stool samples for −80 °C biobanking; performed PCR.
- Cleaned and analyzed microbiome, blood, 24-h blood pressure, and dietary data in R and Python using multivariate and regression methods.
- Coordinated an IRB-approved human biobank study collecting blood and stool samples with health information.

Systematic review: dietary pulses and human health

- Conducted a PICOS-based systematic review of intervention trials; screened 1,800+ records from three databases with machine-learning tools (ASReview, TERA-Tool).
- Applied a large language model (Gemini) to assist data extraction, with every extracted value verified by two human reviewers.

Undergraduate Research Assistant, Xiao Lab 04/2022 – 06/2023
College of Food Science and Engineering, Northwest A&F University · Advisor: Dr. Chunxia Xiao

- Led an independent project on how three lactic acid bacteria species change the phenolic content, antioxidant activity, and starch digestibility of proso millet (National Undergraduate Life Sciences Research, Silver Prize).
- Performed spectrophotometric color analyses of gluten-free proso millet bran cake (co-author, *Foods* 2023).
- Synthesized 50+ articles on coarse grain consumption and type 2 diabetes (co-author, *J Sci Food Agric* 2023).

PUBLICATIONS

co-first author; * corresponding author

In preparation

- **Tang J**#, Adinchezor C#, Grimaldo C, Zhang Y, Iwuamadi I, Rasmussen C, Rose D, Majumder K, Deehan E*. Exploring the impact of different dietary pulse market classes on human gut microbiota and health.
- **Tang J**#, Rasmussen C#, Deehan E*. Effect of distinct dietary pulse market classes on gut microbiome and health markers: a systematic review of intervention trials.
- Grimaldo C, **Tang J**, Whisenhunt L, Chai W, Deehan E*. Evaluating interrelationships between body composition and immunometabolic dysfunction using bioelectrical impedance analysis: an observational study.

Peer-reviewed

- Xiao J, Li Y, Niu L, Chen R, **Tang J**, Tong Z, Xiao C*. Effect of adding fermented proso millet bran dietary fiber on micro-structural, physicochemical, and digestive properties of gluten-free proso millet-based dough and cake. *Foods*. 2023;12(15):2964.
- Deng X, Niu L, Xiao J, Guo Q, Liang J, **Tang J**, Liu X, Xiao C*. Involvement of intestinal flora and miRNA into the mechanism of coarse grains improving type 2 diabetes: an overview. *J Sci Food Agric*. 2023;103(9):4257–4267.

PRESENTATIONS

- **Tang J**. Exploring the impact of different dietary pulse market classes on human gut microbiota and health. Nebraska Food Innovation Center, 05/2026. (Talk)
- **Tang J**, Grimaldo C, Whisenhunt L, Rasmussen C, Majumder K, Rose D, Deehan E. Effect of different Nebraska dry bean market classes on human health and the gut microbiota: a 3-phase randomized crossover clinical trial. Missouri Valley Branch ASM Annual Meeting, 03/2026. (Poster)
- Rasmussen C, **Tang J**, Grimaldo C, Benson A, Rose D, Mukhopadhyay I, Majumder K, Deehan E. Effect of different Nebraska dry bean market classes on human health and the gut microbiota: a preliminary analysis. Nebraska Research Days, 04/2026. (Poster)
- Rasmussen C, **Tang J**, Deehan E. Dietary pulse market class-specific effects on gut microbiota, metabolites, and human health: a systematic review. Nebraska Research Days, 04/2026. (Poster)
- **Tang J**, Whisenhunt L, Rose D, Majumder K, Deehan E. Exploring the impact of different dietary pulse market classes on host metabolic health and gut microbiome. UNL Food Science and Technology Symposium; Nebraska Research Days; NPOD 16th Annual Spring Retreat, 04/2025. (Posters)

AWARDS AND FUNDING

- Chancellor's Graduate Research Grant Award, UNL ($400) 2026
- Departmental Graduate Research Assistantship, UNL ($25,500; $24,500) 2024 – 2026
- Excellence in the Department of Food Science and Technology, UNL 2025
- Mary and Charles C. Cooper / Emma I. Sharpless Fellowship, UNL ($3,000) 2024
- National Undergraduate Life Sciences Research Competition, Silver Prize (¥3,000) 2023
- Undergraduate Merit Scholarship, NWAFU (three years) 2020 – 2023

TEACHING AND MENTORING

Teaching Assistant, FDST 458 Advanced Food Analysis, UNL 01/2026 – 05/2026

- Prepared lab materials, led lab sessions, and graded assignments and exams for 35 students.

Guest Lecturer, FDST 498 Undergraduate Research Experience, UNL 02/2025 – 04/2025

- Delivered three lectures on systematic reviews: search strategy, PICOS design, and AI/ML tools for screening and extraction.

Teaching Assistant, FDST 412, FDST 429, and NUTR 250, UNL 08/2023 – 05/2024

- Graded assignments and proctored exams for 55 students.

Research Mentor, UNL and NWAFU 2023 – 2026

- Co-supervised an undergraduate researcher on a systematic review, trial data entry, and research ethics (UNL, 08/2025 – 05/2026).
- Mentored an undergraduate in literature review (UNL, 2025), a new M.S. student (UNL, 2025 – 2026), and two undergraduates in lab techniques (NWAFU, 2023).

LEADERSHIP AND SERVICE

Graduate Representative, Graduate Student Assembly, UNL 09/2025 – 06/2026

Graduate Student Advisory Board Member, Food Science & Technology, UNL 08/2024 – 08/2025

- Founded the Graduate Student Mentoring Program, pairing each incoming graduate student with a mentor.

Graduate Representative, College of Agricultural Sciences & Natural Resources, UNL 08/2024 – 05/2025

- Organized three professional development workshops on faculty careers, the academic job market, and CV writing.

Junior Reviewer, *Clinical Nutrition* 11/2024

Academic Advisor and Coordinator, Food Science & Engineering, NWAFU 10/2021 – 05/2024

- Advised 19 first-year students and served as liaison for 59 students in the "3+1" program.

Communications Lead, College of Food Science and Engineering, NWAFU 09/2020 – 06/2022

- Led a 13-member team producing weekly newsletters and videos.

SKILLS

- **Computational antibody modeling:** AlphaFold/ColabFold structure prediction; framework grafting; antibody-design tools (RFantibody, Germinal); structural and interface evaluation; high-performance computing; AI coding agents (Claude Code, Codex) for building and validating workflows
- **Programming and statistics:** Python, R, SAS; experimental design and power analysis; multiple regression and multivariate analysis; interpretation in biological and clinical context
- **Wet lab:** PCR; human blood and stool processing (centrifugation, plasma/serum separation, homogenization, aliquoting); −80 °C biobanking; in vitro fermentation; spectrophotometric assays
- **Clinical research:** protocol and SOP development; IRB/GCP procedures; ClinicalTrials.gov registration; participant recruitment and visit coordination; grant proposal writing
- **Evidence synthesis:** systematic review design (PICOS), ML-assisted screening, LLM-assisted extraction with human validation
- **Languages:** Mandarin (native), English (fluent)

TRAINING, CERTIFICATIONS, AND MEMBERSHIPS

- CITI Program: GCP for Clinical Trials with Investigational Drugs and Biologics (2024); Biomedical Responsible Conduct of Research (2024); Biomedical Investigators and Key Personnel (2023); Conflicts of Interest (2023)
- Python for Data Science, AI & Development, Coursera (2025); Data Science: R Basics, HarvardX (2025)
- High Performance Computing for Researchers, Holland Computing Center, UNL (2025); Grant Writing Seminar, UNL (2025)
- Seminar series: Complex Biosystems Seminar Series (2025 – 2026) and Nebraska Food for Health Center Seminars (2024 – 2026), UNL
- Memberships: American Society for Microbiology (2024 – 2026; Missouri Valley Branch 2025 – 2026) · American Society for Nutrition (2024 – 2026) · ISAPP Students and Fellows Association (2023 – 2025)